SCAN CHAIN DIAGNOSIS AGENT

Real time visibility into scan test results for faster debug and higher quality

WHAT DOES THE AGENT DO?

The agent automatically processes chip scan-test fail logs and visualizes diagnostics on a dedicated dashboard. It pinpoints:

- Failed chains & broken scan lines
- Suspect bad cells & physical probe coordinates
- Shift vs. capture failure analysis
- Priority chains to analyze first & confidence scores

Beyond localization, the integrated ML model predicts the likely defect type and root cause of the failure.

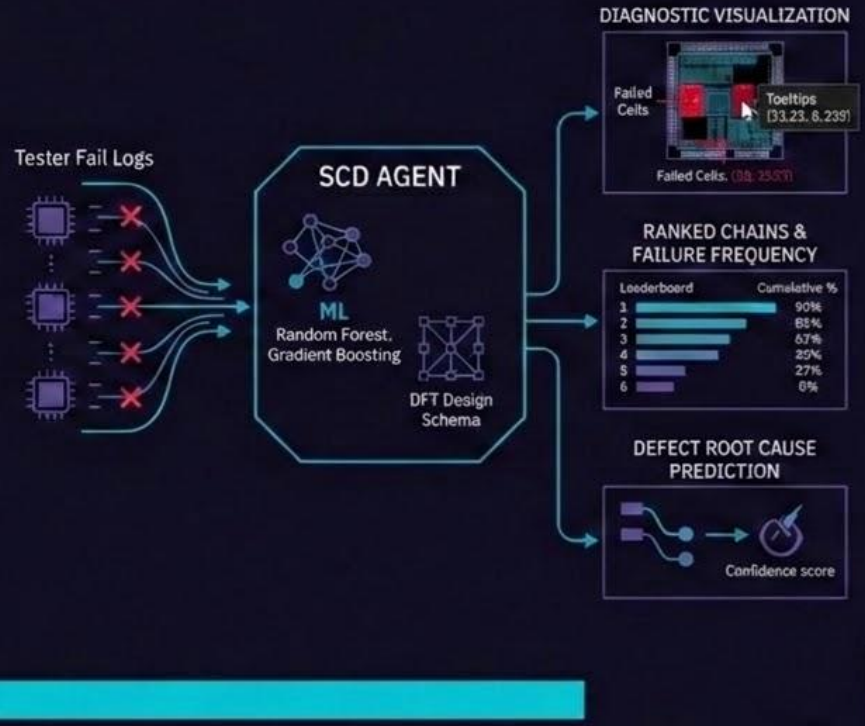


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SCD-FR-001 IDENTIFYING FAILING SCAN CHAINS

Requirement: Detect & list which scan chains failed.

Input/Parse



Raw Tester Log

Locate & extract FAIL lines

Structure/Group



Localized Fault Areas

Group Failures by Chain Name, structuring data into localized areas

Ranking/Prioritize



Prioritized Debug List

Rank Scan Chains based on failure frequency

SCD-FR-002 LOCATING FAILING SCAN CELLS

Requirement: Map & rank failures to physical cells.



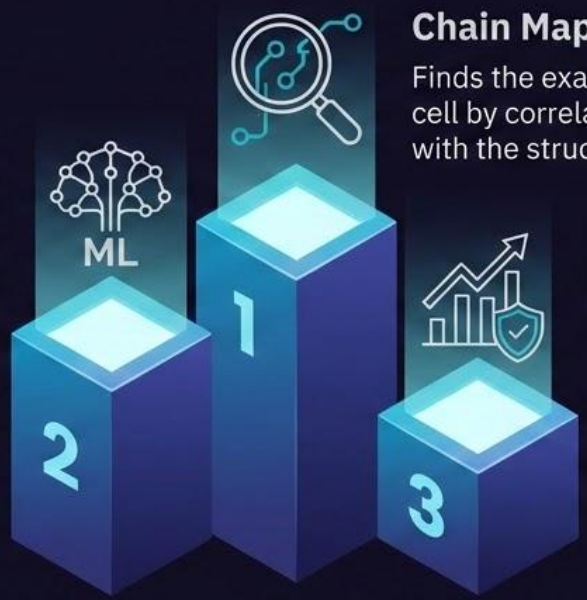
Raw Tester Log



Cell Map Correlation

Cause Prediction

Employs a **Random Forest** machine learning model to predict the underlying cause of the failure.



Chain Mapping

Finds the exact failing physical cell by correlating raw tester flops with the structural scan-chain map.

Confidence Score

Utilizes calibrated **Gradient Boosting** to compute and set the precise confidence percentage.

This step translates raw tester logs into actionable circuit locations for physical debug.

SCD-FR-003 SCAN TOPOLOGY ANALYSIS

Requirement: Build the design map of the scan chains.

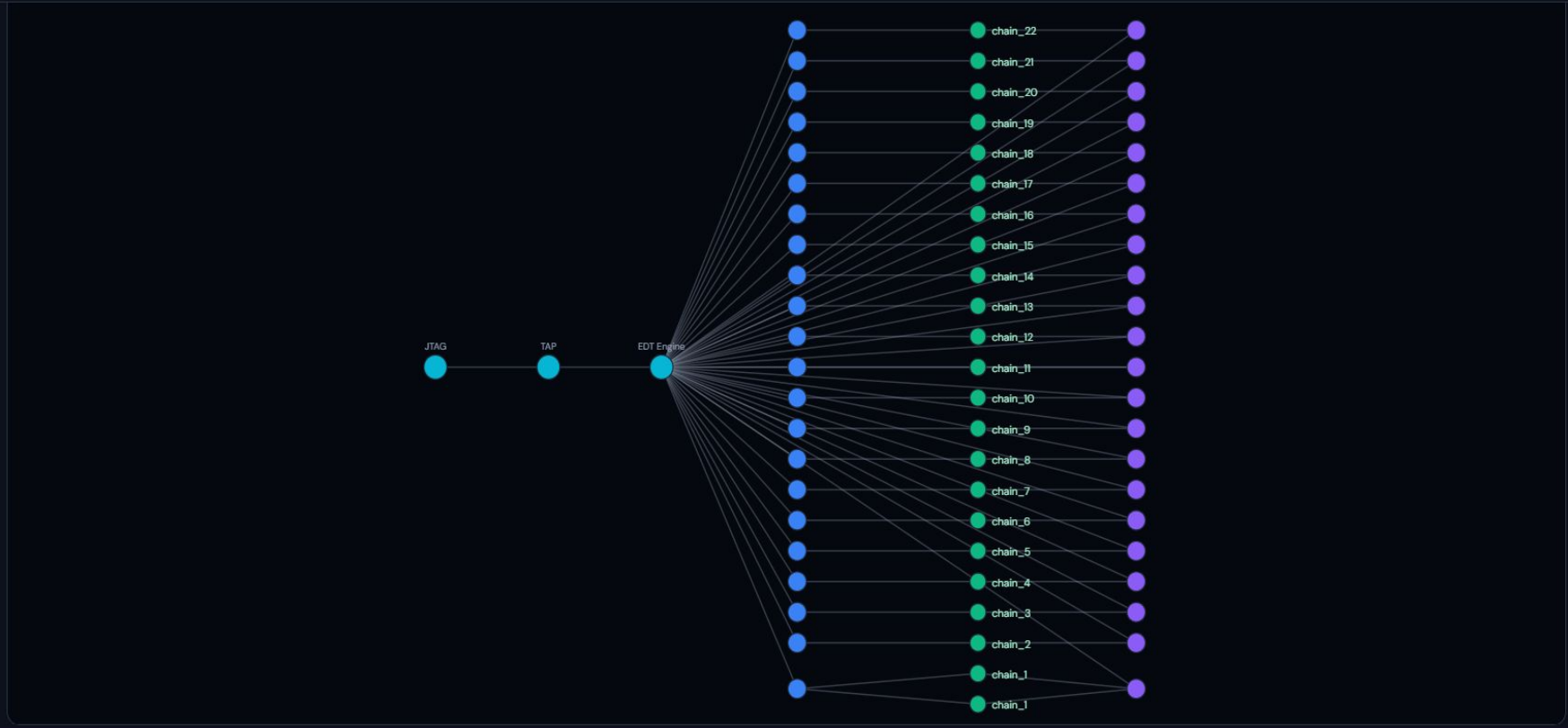
FR-003 reads the DFT design schema to construct a comprehensive scan-chain wiring map.

This topology structure registers crucial physical metrics, including exact **chain lengths**, precise **cell ordering**, clock domains, and EDT pins.

By mapping out exactly how cells are wired together, later diagnostic steps can successfully correlate raw logical errors to exact physical hardware locations.

KPI WORKSPACE

Chains in Topology



SCD-FR-004 FAILURE FREQUENCY RANKING

Requirement: Rank failing scan chains to prioritize engineering debug.



FAIL

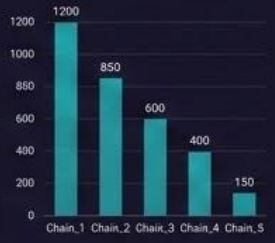
Identify failing scan chains (FR-001)

01

Identify failing scan chains (FR-001)

02

Count Failures



03

Calculate Rates



04

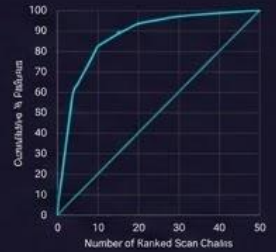
Dense Rank





05

Cumulative %



FR-005 — Correlate Failures

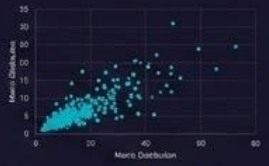
Requirement: Analyze relationship between scan chain failures and physical/topology metrics.

This feature correlates failing scan chains (from **FR-001**) and topology (from **FR-003**) against 18 physical, timing, scan, spatial, and topology metrics to identify root causes.

01

Physical & Timing

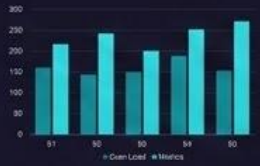
IR Drop, Temperature, Setup Slack, Hold Slack, AI Severity Score.



02

Scan Load

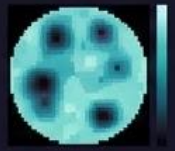
Shift Cycles, Capture Cycles, Scan Fail Count, Transition Faults, Test Time.



03

Spatial Metrics

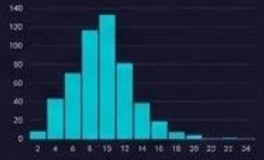
Die Row, Die Column, Wafer X, Wafer Y.



04

Topology

Scan Length, Instance Type, Compression Channel Count.



05

Location

Failure Region, Die Label, Die Row, or Defect Type.



FR-006 — Identify Scan Chain Breaks

Requirement: Locate shift path failures by comparing expected and actual bitstreams.

01

Input Data

Expected Output, Actual Output, and Cell Map



02

Filtering

Remove trailig data comparison data.



03

Correlation



Combine mismatch locations across patterns to identify the most likely break bit/cell.

04

Mapping

Bit Position → Scan Cell Name

Map detected bit position to the corresponding scan cell name using STIL cell order.

05

Confidence

CONFIDENCE ≥ 70?

YES

NO

Output Report

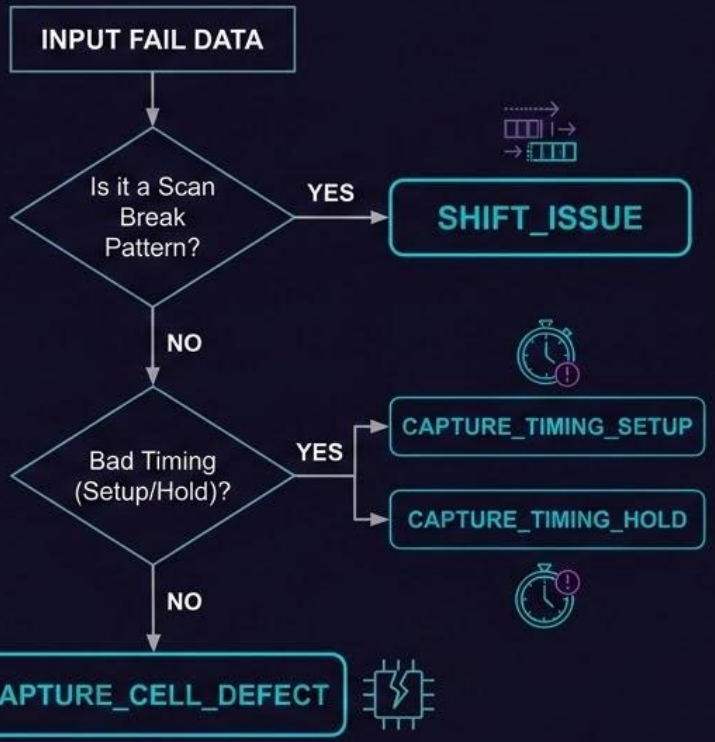
CERTAIN LOCATION
(high confidence, multiple patterns)

UNCERTAIN CANDIDATE
(low evidence)

FR-007 — Detect Shift and Capture Issues

Classify failures and determine the correct debug path.

1. **DETECT SHIFT ISSUES:** Identify failure patterns spanning multiple bit positions. Label as **SHIFT_ISSUE**.
2. **DETECT CAPTURE ISSUES (Timing):** Check for bad setup or hold times. Label as **CAPTURE_TIMING_SETUP** or **CAPTURE_TIMING_HOLD**.
3. **DETECT CAPTURE ISSUES (Defect):** If no shift or timing issues, classify as **CAPTURE_CELL_DEFECT**.



METHODOLOGY

- Isolation Forest for unusual timing
- Random Forest for failure details

FR-008 — Generate Scan Diagnosis Report

Consolidates outputs from all diagnostic modules into a unified, shareable HTML report with embedded visualizations.



01 / DATA INTEGRATION

Multi-Feature Aggregation

Collects and merges diagnostic outputs from **FR-001 to FR-007** and **FR-009** into one unified layout.



02 / CONFIDENCE METRICS

Adaptive Break Resolution

CERTAIN maps the exact faulty cell. **LOCATION_UNCERTAIN** isolates candidate target locations.

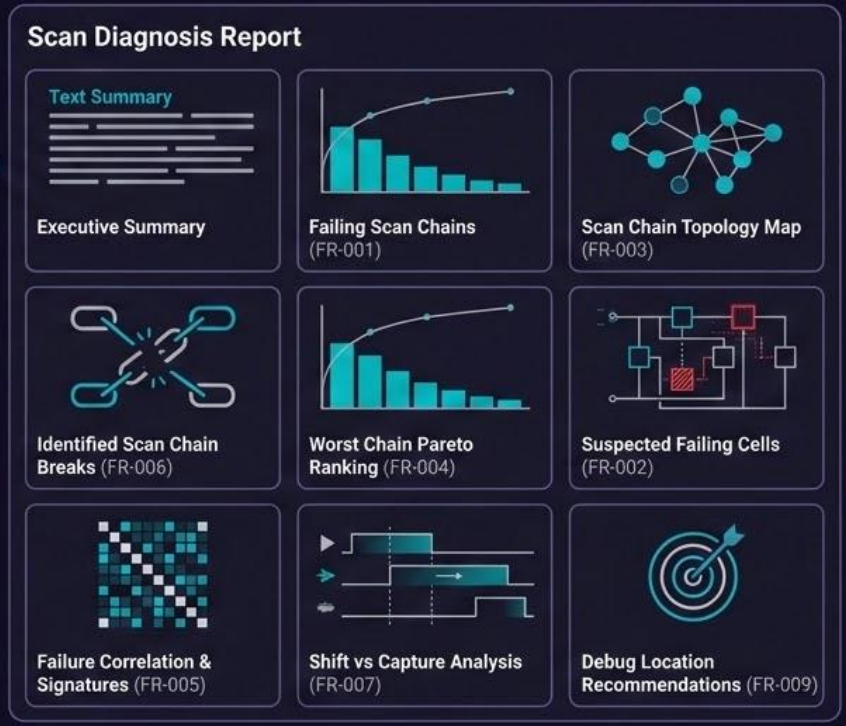


03 / SYSTEM OUTPUT

Deliverable & Metrics Tracking

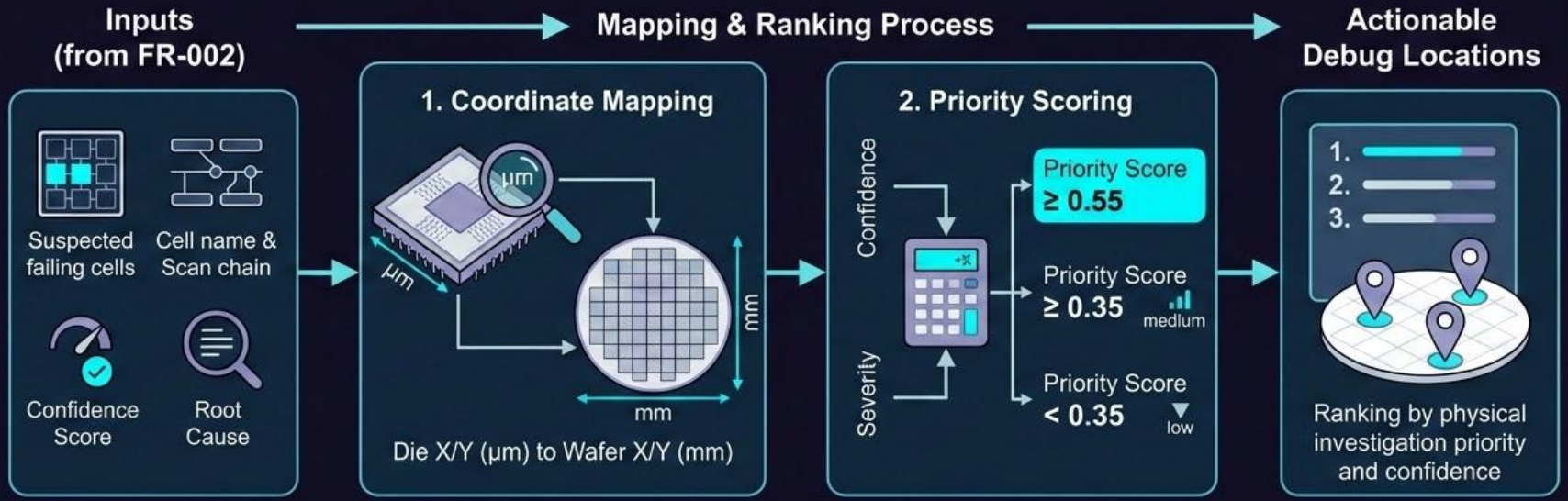
Generates a standalone interactive report and updates KPI counters:

`output/SCD-FR-008_scan_diagnosis_report.html`



SCD-FR-009 — Recommend Debug Locations

This feature identifies the probable physical locations of failures and provides recommended debug points for further analysis.



FR-010 — Diagnosis Confidence

This feature calculates the confidence level of the diagnosis, showing how much the identified failure location can be trusted.

Calculate Evidence Score

- Relative dominance of failing cell compared to others in same chain.
- Agreement of multiple failing patterns.
- Percentage of failures linked to that cell.
- Consistency of the failure type.

Determine ML Probability

- ML model (Gradient Boosting with calibration) to predict probability.
- Note: Logistic Regression fallback if ML model unavailable.

Confidence = 50% Evidence Score + 50% ML Probability

Generate overall **Diagnosis Confidence KPI** using fail-weighted average of top suspected cells from each scan chain.

